U-CMG: U-Shaped Cross-Domain Control Moment Gyroscope Bearing Fault Diagnosis Method

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***Abstract*—To address the problems of insufficient labeled data and distribution differences in single-frame spacecraft control moment gyroscope (CMG) bearing fault diagnosis, a transfer learning method for domain-adaptive CMG bearing fault diagnosis based on the U-Net network structure is proposed. This method improves label reliability and model stability in the cross-domain transfer process through a multi-label adversarial pseudo-label generation strategy. In addition, a feature extractor based on the U-Net network is designed to deeply mine discriminative features of bearing vibration signals, enhancing the model's fault recognition capability. During the entire training process, model parameters are updated through dynamic calibration of pseudo-labels, ensuring that high-confidence data continuously participate in training, thereby improving the model's robustness and generalization ability. Experimental results show that this method achieves an average fault diagnosis accuracy of 92.32% on the spacecraft CMG bearing fault dataset, outperforming four other mainstream transfer learning methods, which verifies the effectiveness and superiority of the proposed approach.**

Keywords—CMG, fault diagnosis, domain adaptation, transfer learning

I. Introduction

The Control Moment Gyroscope (CMG) is a core actuator in spacecraft attitude control systems, and its performance directly affects spacecraft attitude stability and mission reliability[1]. Among its components, bearings serve as key parts that operate at high speed and over long lifetimes. During prolonged operation, bearings are susceptible to frictional wear, material fatigue, and lubrication degradation, which can lead to failures[2]. Once a bearing fails, the CMG may produce abnormal torque, further causing a decline in overall spacecraft attitude control performance or even mission failure. Therefore, research on CMG bearing fault diagnosis has significant engineering and theoretical value.

With the development of sensors and intelligent diagnostic technologies, deep learning-based fault diagnosis methods have been widely studied. These methods can automatically extract high-dimensional features from vibration signals and generally achieve high diagnostic accuracy when labeled data are sufficient and distributions are consistent. However, in practical spacecraft applications, the number of bearing fault samples is often limited due to testing conditions and mission environments, and cross-condition data exhibit significant distribution differences. Directly applying traditional deep learning models in such scenarios often leads to severe performance degradation. To address this issue, researchers introduce transfer learning and domain adaptation methods to enhance model generalization through cross-domain knowledge transfer. Although existing studies partially improve cross-domain diagnostic performance, they still face two major challenges: first, the source and target domain distributions may differ too much, leading to insufficient feature transfer; second, noise may be introduced during pseudo-label generation, reducing the stability and reliability of model training.

To tackle these problems, this paper proposes a transfer learning method based on a domain-adaptive teacher model. The main contributions are as follows:

- Cascaded Pseudo-labels: A pseudo-label generation strategy that effectively combines local and global features to generate multi-level pseudo-labels. It utilizes adversarial learning with pseudo-labels to achieve target domain adaptation, improving the reliability of cross-domain pseudo-labels and reducing interference from incorrect labels during model training.
- Feature extractor based on U-Net network: Designed to fully exploit the discriminative features of bearing vibration signals and enhance cross-domain diagnostic capability.
- Dynamic calibration mechanism during training: Achieves cross-domain feature alignment and knowledge transfer. Experimental results show that the proposed method significantly outperforms various comparison methods in average recognition accuracy on the spacecraft CMG bearing fault dataset, demonstrating its effectiveness and superiority.

II. Related Theories

A. Unsupervised Domain Adaptation

In unsupervised domain adaptation (UDA), DANN is one of the representative methods. Its core idea is to use adversarial training to enable the feature encoder to learn domain-invariant representations, thereby reducing the distribution discrepancy

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between the source domain and the target domain. DANN mainly consists of a feature encoder, a task classifier, and a domain classifier[3]. Specifically:

- Feature encoder: maps the input into a latent feature space.
- Task classifier: trains on labeled data from the source domain to perform the main task.
- Domain classifier: distinguishes whether the features come from the source domain or the target domain.

During training, the output of the feature encoder is fed simultaneously into the task classifier and the domain classifier. The task classifier optimizes the task accuracy on the source domain, while the feature encoder and domain classifier establish an adversarial relationship through a gradient reversal layer (GRL): the domain classifier attempts to distinguish the domain origin, whereas the feature encoder tries to generate features that are difficult to distinguish, achieving cross-domain alignment. The final optimization objective is a joint loss: minimizing the source domain task loss while maximizing the domain classifier error. This mechanism aligns cross-domain features while maintaining discriminability and has demonstrated remarkable performance in image recognition, speech processing, and mechanical fault diagnosis[4].

B. *Multi-Head Broadcast Self-Attention*

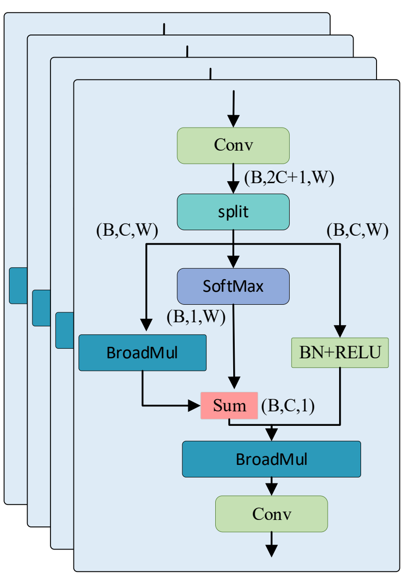

Fig. 1. Network Structure of the Multi-Head Broadcast Self-Attention

The multi-head self-attention module, as a core component of the Transformer, captures multiple types of relationships in the input data through parallel computation. Each element attends to and weights other elements to generate new outputs, without relying on external information, and it is widely used in fields such as natural language processing.

The computation process first obtains Q, K, and V through linear transformations, then splits them into multiple "heads" for self-attention computation. The results are concatenated and integrated through a linear transformation. Using the broadcast mechanism in the PyTorch framework, the module traverses temporal information in one dimension, reducing computational complexity while fully utilizing global information[5]. The network structure of the multi-head broadcast self-attention mechanism is shown in Fig. 1. Its computation process is as follows:

$$\begin{cases} Q = SoftMax\left(XW_Q\right), K = XW_K, V = RELU\left(XW_V\right) \\ W_Q \in \mathbb{R}^{C_H*1}, W_K, W_V \in \mathbb{R}^{C_H*C} \\ Q = \left(Q_1, Q_2, \ldots, Q_h\right), K = \left(K_1, K_2, \ldots, K_h\right), V = \left(V_1, V_2, \ldots, V_h\right) \\ O_i = Attention\left(Q_i, K_i, V_i\right) = \left(Sum\left(Q_i \otimes K_i\right)\right) \otimes V_i \\ X' = Concat\left(O_1, O_2, \ldots, O_h\right) W_O \end{cases} \quad (1)$$

Here, $\otimes$ represents the broadcasting operation.

III. METHOD

A. *Overview*

The U-CNG network architecture is shown in Fig. 2. We adopt a symmetric skip-connection design based on the U-Net encoder-decoder structure. The encoder combines the CNN-based SMAConv with the Transformer-based Bottleneck. SMAConv employs large-kernel depthwise separable multi-scale convolutions and an inverted bottleneck structure, which maintains lightweight design while expanding the receptive field and extracting global features. The Bottleneck optimizes global feature interaction and efficiently captures global dependencies. The decoder uses a cascaded stage-wise enhanced sampling structure and improves the reliability of pseudo-labels through multi-level label prediction for adversarial pseudo-label generation.

B. *SMAConv Block*

In practical acquisition, bearing fault signals are usually accompanied by a large amount of noise, which may originate from measurement errors of the sensors themselves, random external disturbances, or coupling effects from other mechanical components. These complex noise characteristics not only obscure fault features but also reduce the ability of traditional methods to recognize fault patterns. Existing depthwise separable convolution structures are mainly designed for natural image tasks, and their convolution kernels have relatively fixed receptive fields, making it difficult to fully capture the multi-scale non-stationary features and long-range dependencies present in bearing fault signals. Therefore, relying solely on such structures cannot robustly extract fault features. To address this problem, we propose a separable multi-scale adaptive convolution module. The module is designed to integrate multi-scale convolution operations with an adaptive feature selection mechanism and employs two inverted pointwise convolutions to effectively model long-range dependencies. On one hand, multi-scale convolutions enhance the model's ability to perceive features at different frequencies and time scales, thereby better covering the multi-level variation patterns of fault signals. On the other hand, an adaptive weight allocation strategy is introduced, allowing the model to dynamically adjust the contribution of convolution

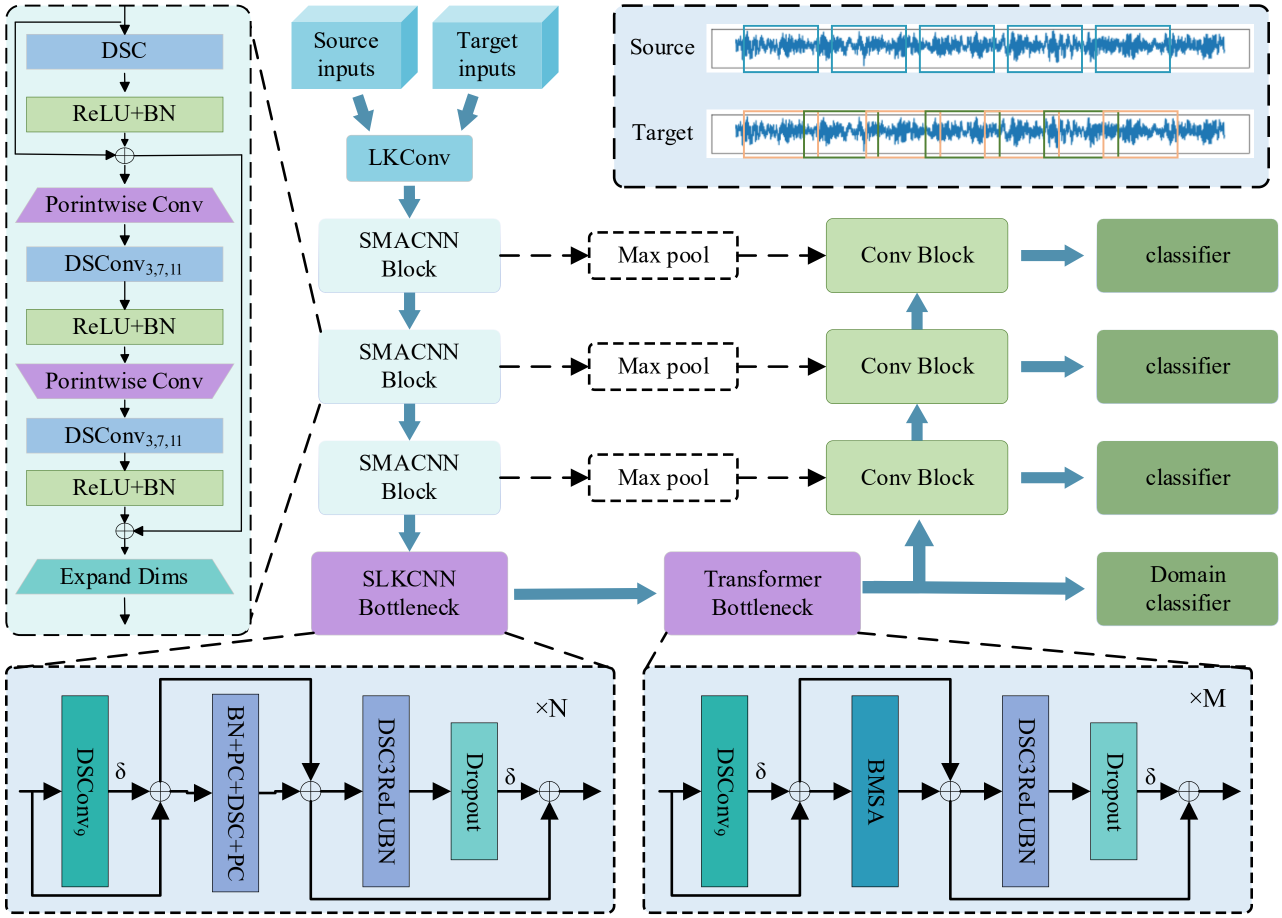

Fig. 2. The architecture of U-CMG

kernels at different scales according to signal characteristics, effectively enhancing the response to key fault features while suppressing irrelevant noise interference. The SMAConv module is defined as follows:

$$
\begin{aligned}
X_1 &= BN\left(f\left(DepthwiseConv\left(X\right)\right)\right)+X\\
y_i^{kl} &= DSC_{kl}\left(PorintwiseConv\left(X_i\right)\right)\\
X_2 &= BN\left(f\left(y_1^{k3}, y_1^{k3}, y_1^{k11}\right)\right)\\
Y &= BN\left(f\left(y_2^{k3}, y_2^{k3}, y_2^{k11}\right)\right)+X_2
\end{aligned}
\tag{2}
$$

Among them, X_i represents the output variable of the intermediate unit, $f(\bullet)$ denotes the ReLU activation function, BN represents batch normalization, l in DSC_{kl} indicates the size of the convolution kernel, l in y_i^{kl} represents the convolution kernel size, and i represents the output variable of the i-th intermediate unit.

C. *Cascaded Pseudo-labels*

In cross-domain transfer learning, pseudo-labels serve as the supervisory information for unlabeled target domain samples and play a crucial role in enhancing the model's adaptability. However, traditional pseudo-label generation methods usually rely on a single prediction, which is easily affected by noise and domain discrepancies, leading to insufficient pseudo-label accuracy and even introducing cumulative errors during model training. To address this issue, we propose a Cascaded Pseudo-label Generation Strategy (Cascaded Pseudo-labels). The core idea of this method is to generate multiple candidate labels through multi-level predictions and introduce an adversarial mechanism for pseudo-label selection and correction, thereby balancing local discriminative features and global semantic information. The model performs hierarchical predictions in different feature spaces to obtain multiple candidate labels. These labels emphasize local fine-grained features and global high-level semantic features, forming a multi-view candidate pseudo-label set. Through the pseudo-label adversarial strategy, candidate labels engage in a game between global consistency and local discriminability, thus improving the stability and robustness of the final pseudo-labels. By adopting a cascaded integration, the method gradually fuses low-level local labels with high-level global labels and dynamically adjusts the weight of each level in the final pseudo-label generation, effectively reducing the bias of single-label prediction. This strategy not only improves the reliability of pseudo-labels in the target domain but also suppresses the interference of incorrect labels, thereby

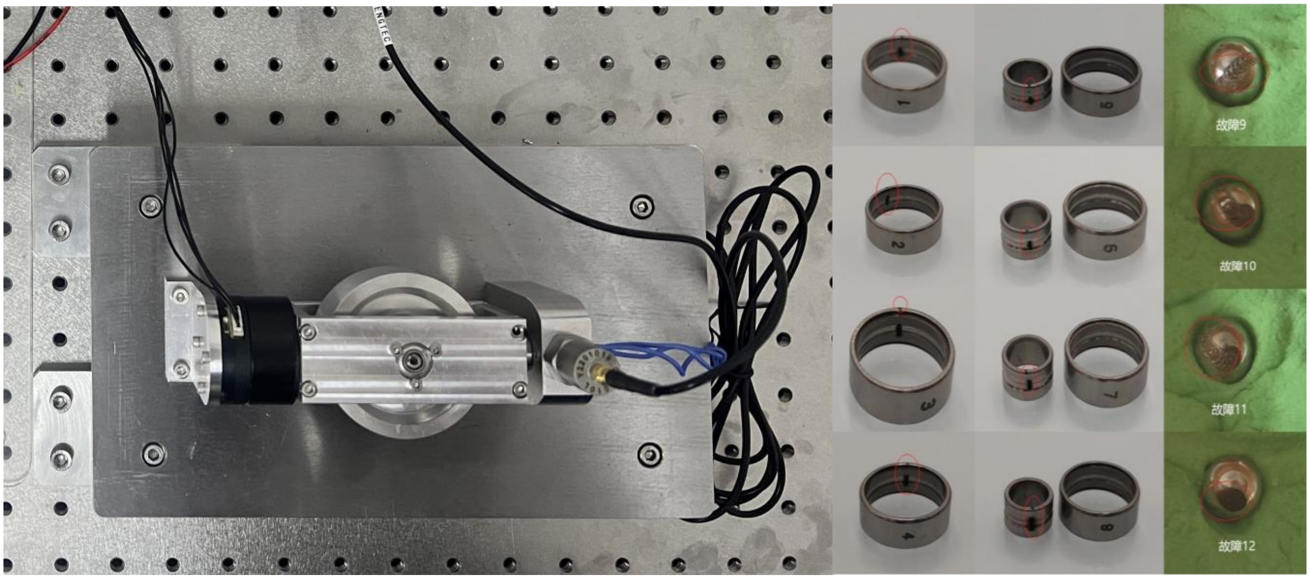

Fig. 3. The test bench of the CMG dataset

significantly enhancing the model's adaptability and generalization performance in cross-domain scenarios.

The cascaded pseudo-label generator produces multi-level candidate labels. The discriminator is trained to distinguish high-confidence labels from low-confidence ones, while the label generator attempts to fool the discriminator. This adversarial game gradually refines pseudo-label quality, ensuring that only high-confidence target samples participate in model updates. Fig. 4 illustrates the iterative selection and refinement procedure.

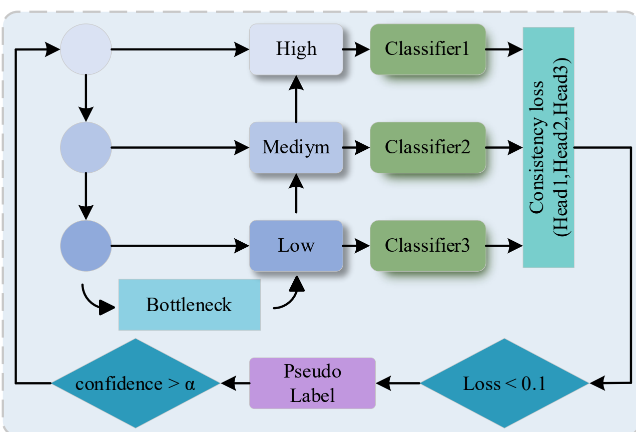

Fig. 4. Logical block diagram of cascaded pseudo-labeling

IV. Experiment Analysis

A. Dataset Description

Three bearing datasets are used to fully verify the proposed method, including the self-built CMG bearing fault dataset A, the CRWU dataset B, and the XJTU bearing dataset C.

Dataset A is a specialized dataset constructed based on the fault characteristics of CMG, and its experimental platform is shown in Fig. 3. The dataset is collected using acceleration sensors and reflects the bearing vibration characteristics under different flywheel weights and typical operating conditions. In four operating conditions, single-point faults are introduced in the rolling element, inner race, and outer race, with fault diameters of 0.1 mm, 0.2 mm, 0.4 mm, and 0.8 mm, simulating progressive damage. The dataset contains a total of 13 operating states (12 fault types plus the healthy state) and covers multiple load conditions (45 g, 55 g, 65 g, and no-load), ensuring broadness and diversity of operating conditions.

Dataset B (CWRU) is collected by the Case Western Reserve University Bearing Data Center. The experimental platform consists of a motor–rotor–bearing system, and single-point faults are introduced in the inner race, outer race, and rolling element through electrical discharge machining, with fault diameters of 0.007, 0.014, and 0.021 inches. The experiments run under different loads (0–3 hp) and speeds, with sampling frequencies of 12 kHz and 48 kHz. The dataset covers normal conditions and multiple fault modes (inner race faults, outer race faults, rolling element faults), and it is widely used for the validation and comparison of bearing fault diagnosis methods.

Dataset C (XJTU_Gearbox) is collected by Xi'an Jiaotong University. The experimental platform consists of a motor, a coupler, a two-stage gearbox, and a loading device. Different types of local defects (such as tooth root cracks, tooth surface pitting, and tooth surface wear) are introduced on the gears to simulate typical gearbox transmission system faults. The experiments run under different speeds and load conditions, with a vibration signal sampling frequency of 20 kHz. The dataset covers normal conditions and multiple gearbox fault modes, and it is widely used for research and validation of gearbox fault diagnosis and health monitoring methods.

TABLE I. ACCURACY RESULTS OF ABLATION EXPERIMEN

method	A→B	A→C	B→A	B→C	C→A	C→B	Mean Accuracy
DAN	65.71	58.96	47.27	78.96	53.12	75.37	63.23
DANN	42.31	54.72	37.51	89.71	43.26	92.32	59.97
DSAN	59.32	73.71	68.31	87.37	73.74	88.73	75.20
JDA	68.72	79.31	78.36	85.32	69.83	89.72	78.54
U-CMG	91.72	85.32	93.46	97.43	90.88	95.11	92.32

B. *Comparative Experiment*

Based on the above three datasets, six transfer tasks are set in the experiments, namely A→B, A→C, B→A, B→C, C→A, and C→B, where "→" denotes transfer from one dataset to another. The baseline methods for comparison are mainstream domain adaptation transfer learning approaches, including Domain Adaptation Network (DAN)[6], Domain-Adversarial Neural Network (DANN)[3], Deep Subdomain Adaptation Network (DSAN)[7], and Joint Distribution Adaptation (JDA)[8]. These methods adopt different strategies to reduce the distribution discrepancy between the source and target domains, thereby addressing the domain adaptation problem. Their common goal is to improve cross-domain performance, while their differences lie in the strategies employed. DAN enhances cross-domain performance by minimizing the feature discrepancy between source and target domains. DSAN further achieves subdomain-level alignment on the basis of DAN, better capturing subtle differences under complex distributions. DANN leverages adversarial training to extract domain-invariant features, thus enhancing cross-domain generalization capability. JDA aligns both marginal and conditional distributions, making it more comprehensive than methods aligning a single distribution when adapting to data characteristics across domains. The experimental results are shown in Tab I.

In the six cross-domain transfer tasks, the classification accuracy of different methods shows significant variation. Traditional methods generally achieve low overall accuracy with large fluctuations. For example, the accuracy of DANN is only 42.31% in the A→B task, while it reaches 92.32% in the C→B task, indicating unstable performance across tasks. In comparison, DAN achieves an average accuracy of 63.23%, DSAN improves it to 75.20% through subdomain alignment, and JDA reaches an average accuracy of 78.54% by aligning both marginal and conditional distributions, outperforming DAN, DANN, and DSAN overall.

The proposed U-CMG model achieves significant advantages across all transfer tasks, with an average accuracy of 92.32%, clearly superior to the compared methods. In the B→A and B→C tasks, the accuracy reaches 93.46% and 97.43%, respectively, demonstrating strong cross-domain adaptability. These results indicate that the SMAConv module, through multi-scale feature modeling and adaptive weight allocation, effectively enhances the model's ability to perceive complex patterns in bearing signals. The cascaded pseudo-label generation strategy improves the stability and reliability of pseudo-labels in the target domain, thereby reducing the interference of incorrect pseudo-labels. The overall network combines the strengths of convolution and Transformer, achieving a better balance between local feature extraction and global dependency modeling. Although U-CMG demonstrates strong transferability, performance may degrade when the target domain exhibits extremely high noise levels or unseen failure modes. In such cases, pseudo-labels generated in early training steps may contain noise, which can influence convergence. Future work will consider incorporating noise-robust confidence estimation and online pseudo-label correction to improve stability in highly uncertain environments.

To evaluate the computational efficiency of U-CMG, we report the number of parameters and floating-point operations (FLOPs) of the proposed model. The SMAConv block introduces multi-scale kernels, but the use of depthwise separable convolution keeps the computational cost lightweight. The overall model contains 0.876M parameters and 2.1 GFLOPs, which is comparable to lightweight domain adaptation models such as DANN, while significantly lower than Transformer-heavy architectures. This ensures that the model can be deployed on embedded or on-board monitoring units where computational and memory resources are limited.

V. EXPERIMENT ANALYSIS

This paper addresses the problem of cross-domain fault diagnosis of spacecraft Control Moment Gyroscope (CMG) bearings under a single-frame condition and proposes a domain adaptation transfer learning method (U-CMG) based on the Unet network structure. By incorporating a cascaded pseudo-label generation strategy with multi-label adversarial learning and a multi-scale adaptive convolution module (SMAConv), the method effectively alleviates the challenges of insufficient labeled data in the target domain and large cross-domain distribution discrepancies.In terms of pseudo-label generation, the cascaded strategy combines multi-level candidate labels with an adversarial selection mechanism, improving the reliability and stability of pseudo-labels in the target domain and reducing the interference of noisy labels during model training. In terms of feature extraction, the SMAConv module leverages multi-scale convolution and adaptive weight allocation to deeply capture complex patterns in bearing vibration signals, effectively suppress noise interference, and balance local fault features with global dependencies. Experimental results show that U-CMG achieves an average fault diagnosis accuracy of 92.32% across multiple cross-domain transfer tasks, significantly outperforming mainstream transfer learning methods such as DAN, DANN, DSAN, and

JDA. These findings verify the effectiveness of the proposed method in cross-domain knowledge transfer and feature alignment. U-CMG not only improves the accuracy and stability of cross-domain fault diagnosis but also provides a new technical perspective for cross-domain diagnosis of complex mechanical systems. Future work can further explore multi-source domain transfer, online adaptive updates, and lightweight model deployment to enhance the applicability of the method in real engineering scenarios.The proposed framework is not limited to CMG bearings and can be extended to other rotating machinery, such as gears, motors, or wind turbines. However, domain-specific fine-tuning may still be required to accommodate different signal distributions.

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